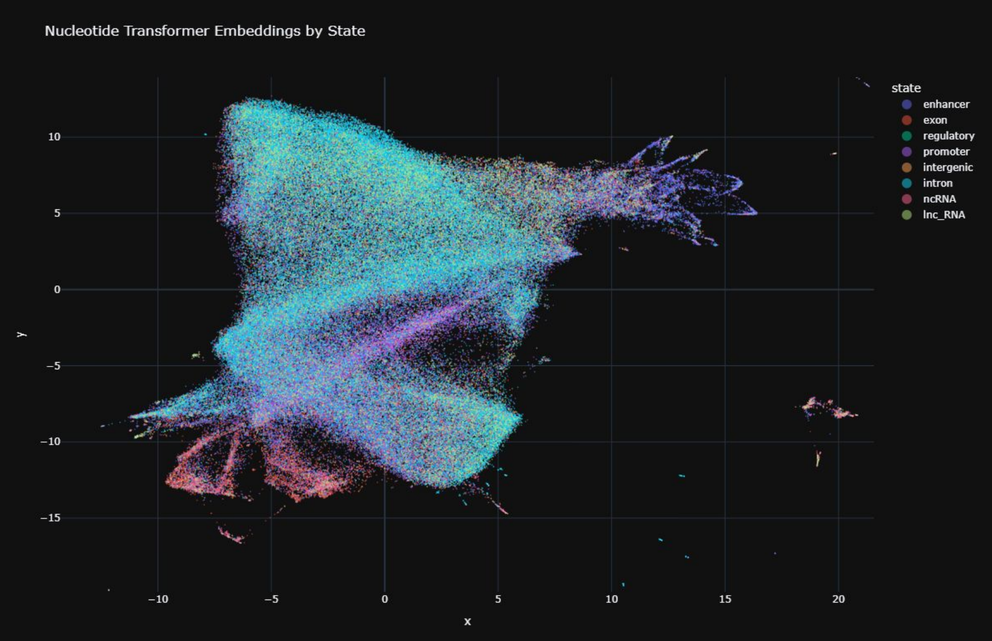
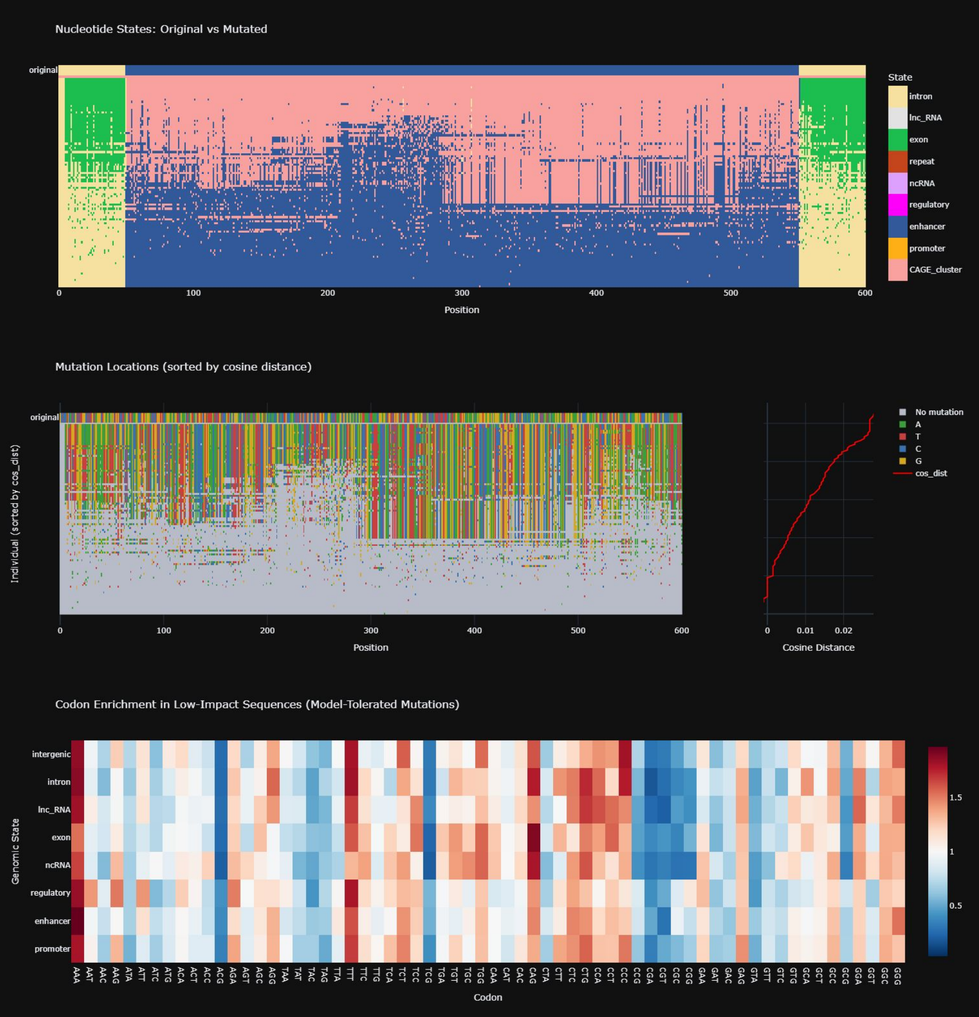
Does Nucleotide Transformer Understand Genomics?

Examining embeddings with stochastic and probabilistic methods

Abstract:

Embedding models are often touted as providing robust representations of meaning in natural language (e.g. king - man = queen - woman). This study uses a genetic algorithm (NSGA-II) and HMM trained on NCBI-GFF annotated regions to evaluate the stability and depth of Nucleotide Transformer's embeddings on genomic sequences. Using human genome HG38, 78 sequences were iteratively modified to generate the data below. Each consisted of a +10% extended window in either direction to capture region transitions. This study finds a high correlation with phyloP scores, indicating the model has developed an understanding of the functional implications of conserved regions.





Mutation arrays:

The genetic algorithm and hidden markov model work together to flip hidden states and nucleotides to **maximally fool** Nucleotide Transformer by choosing mutations it notices least.

State Array: The figure to the left depicts nucleotide hidden states and their corresponding flips. Cosine distance increases from bottom to top, and so does mutation frequency. Enhancer → CAGE cluster, intron → exon.

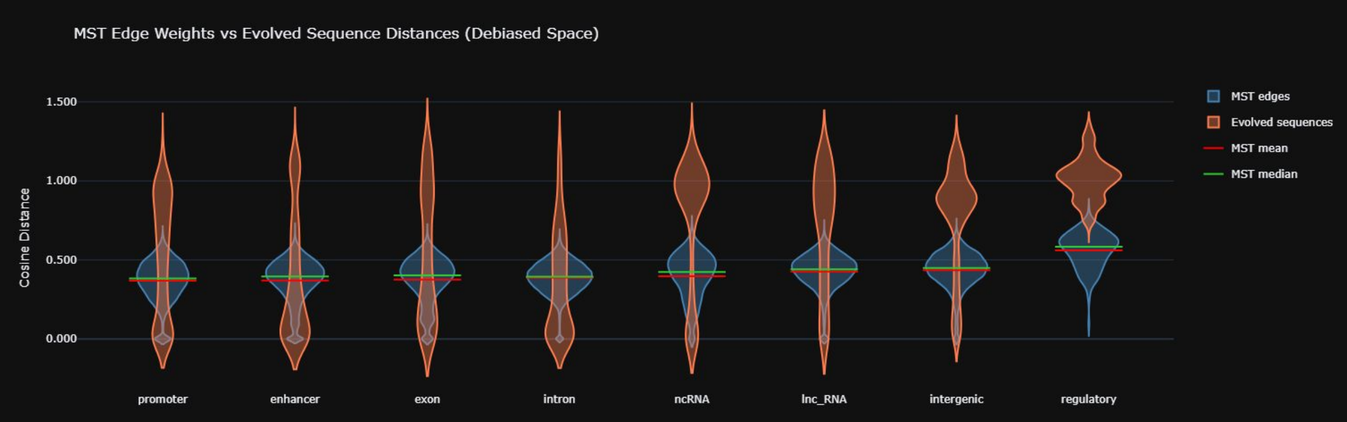
Mutation Array: The middle figure shows which bases were chosen and the corresponding cosine distance. Note the homopolymeric regions.

Codon map: The odds ratio for codons in all mutated sequences relative to originals, normalized by cosine distance. Note the strong preference for homopolymers (e.g. AAA, TTT, CCC, GGG) and avoidance of GC-related codons.

Cosine Distances:

Within-region distance variability (blue) compared to mutated distances (orange).

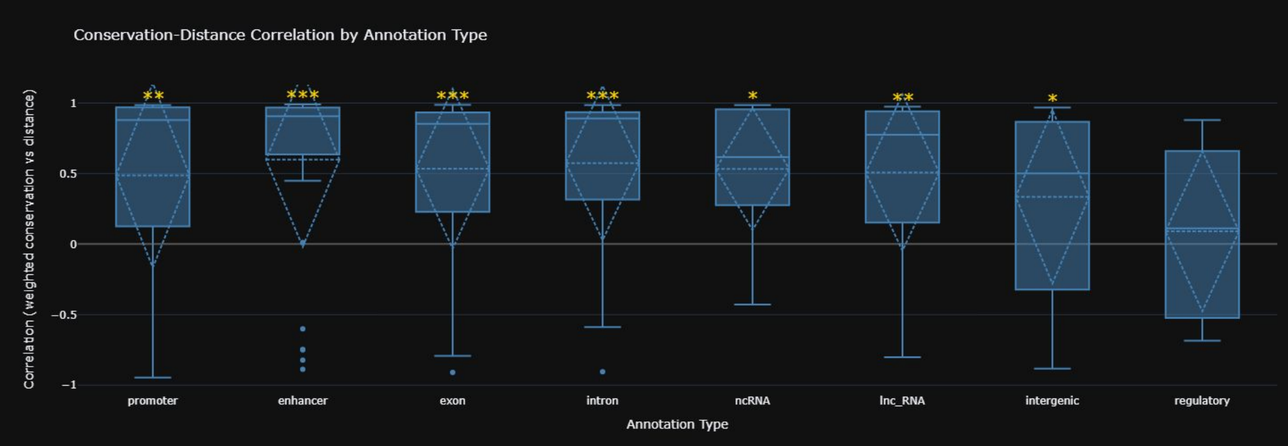
- **Many** mutated sequences fall within average region cosine distances.
- Promoters, enhancers, exons, introns **most susceptible** to mutation.
- Regional cosine distance computed using MST of pairwise cosine distances (k = 15).



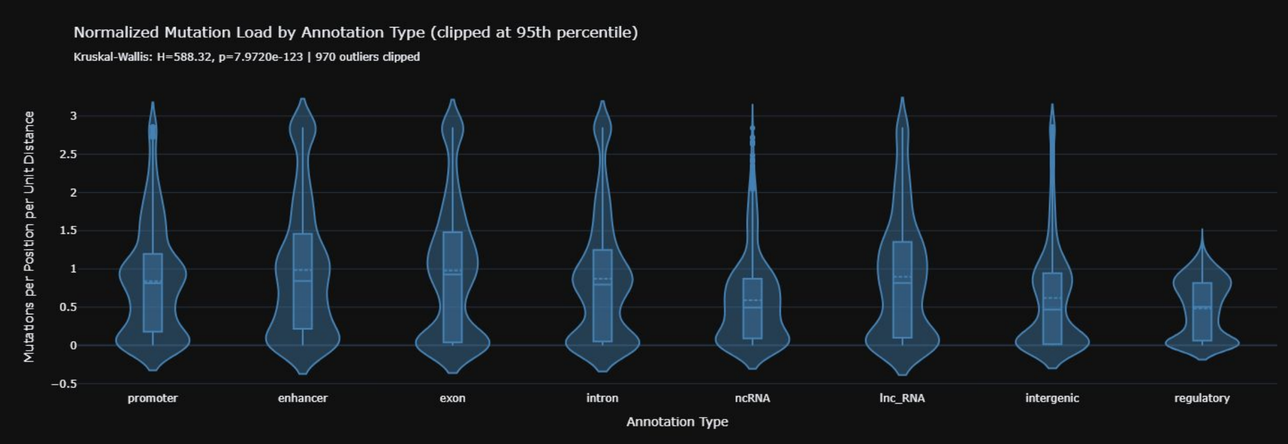
Conservation Scores:

PhyloP scores represent regions that co-occur across species at the nucleotide-level. These are more functionally "essential" and universal.

- **Statistically significant** relationship with cosine distance for most regions.
- Many frames (~60%) have a correlation close to 1.0.



Mutation Load:

The number of mutations per array normalized by length. Weighted inverse cosine distance.

- Normalized mutation load varies significantly across regions.
- Enhancers, exons, introns, lnc_RNA have higher normalized mutation load.